

$$2^{r+w} = 4000 \times 4$$

$$r+w=14 \quad \Rightarrow r=10$$

$$\rightarrow \text{tag} = S - r$$

$$20 - 10$$

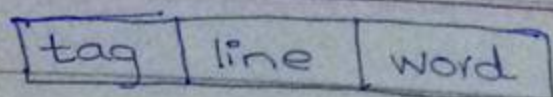
$$\text{line} = \log_2 (\text{cache capacity})^{\text{block}}$$

$$\text{cache capacity} = \frac{\text{cache capacity}}{\text{block size}}$$

$$= \frac{16000}{16}$$

$$\text{line} = \log_2 (1000)$$
$$\approx 10$$

DIRECT MAPPING:



Block size: 2^w bits in word

Blocks of main memory: 2^s

Main memory size = 2^{s+w} Address

Cache memory size = 2^{r+w} length.

$$\text{tag} = s - r$$

$$\text{line} = \log_2(\text{cache block})$$

↳ cache capacity
block size

Mid Q#2:

$$\text{Main memory size} = 2^{s+w} = 16 \times 10^6$$

Ins.

$$\text{Cache capacity} = 4000 \times 4$$

$$\text{Data cache} : 16000 \times 4$$

$$2^w = 16$$

$$\Rightarrow w = 4$$

$$2^{s+w} = 16 \times 10^6$$

$$4 + s = 24 \Rightarrow s = 20$$